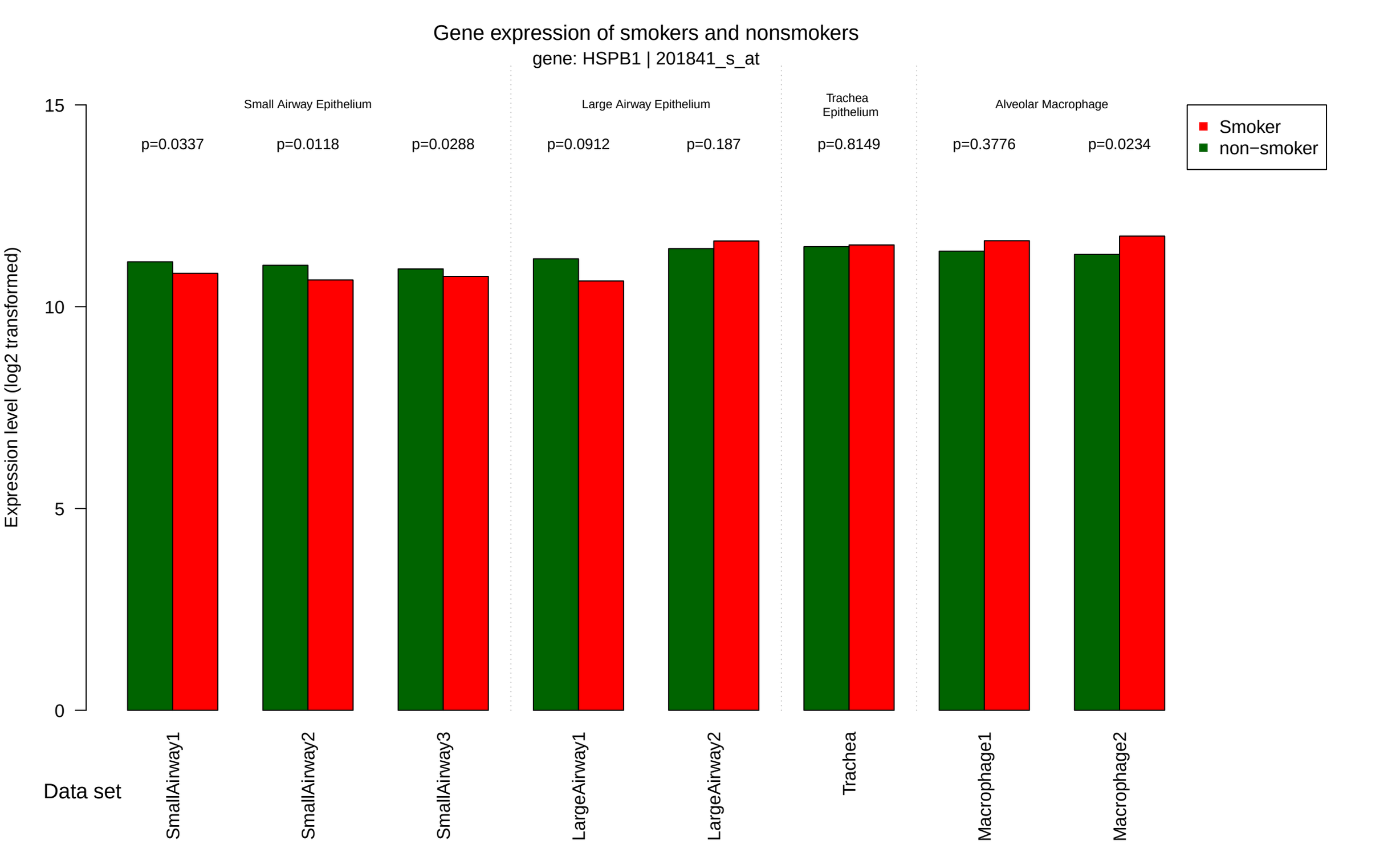
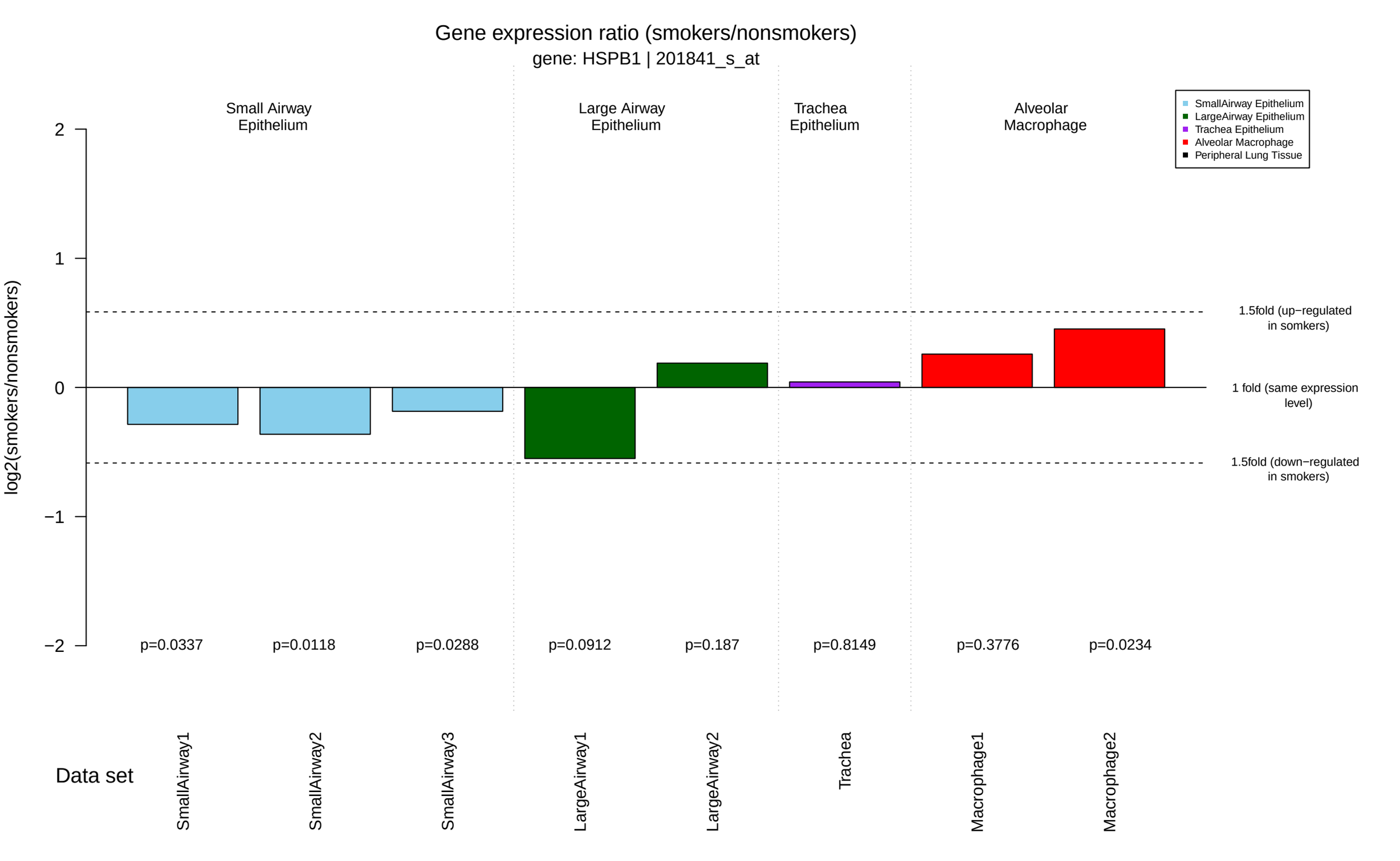
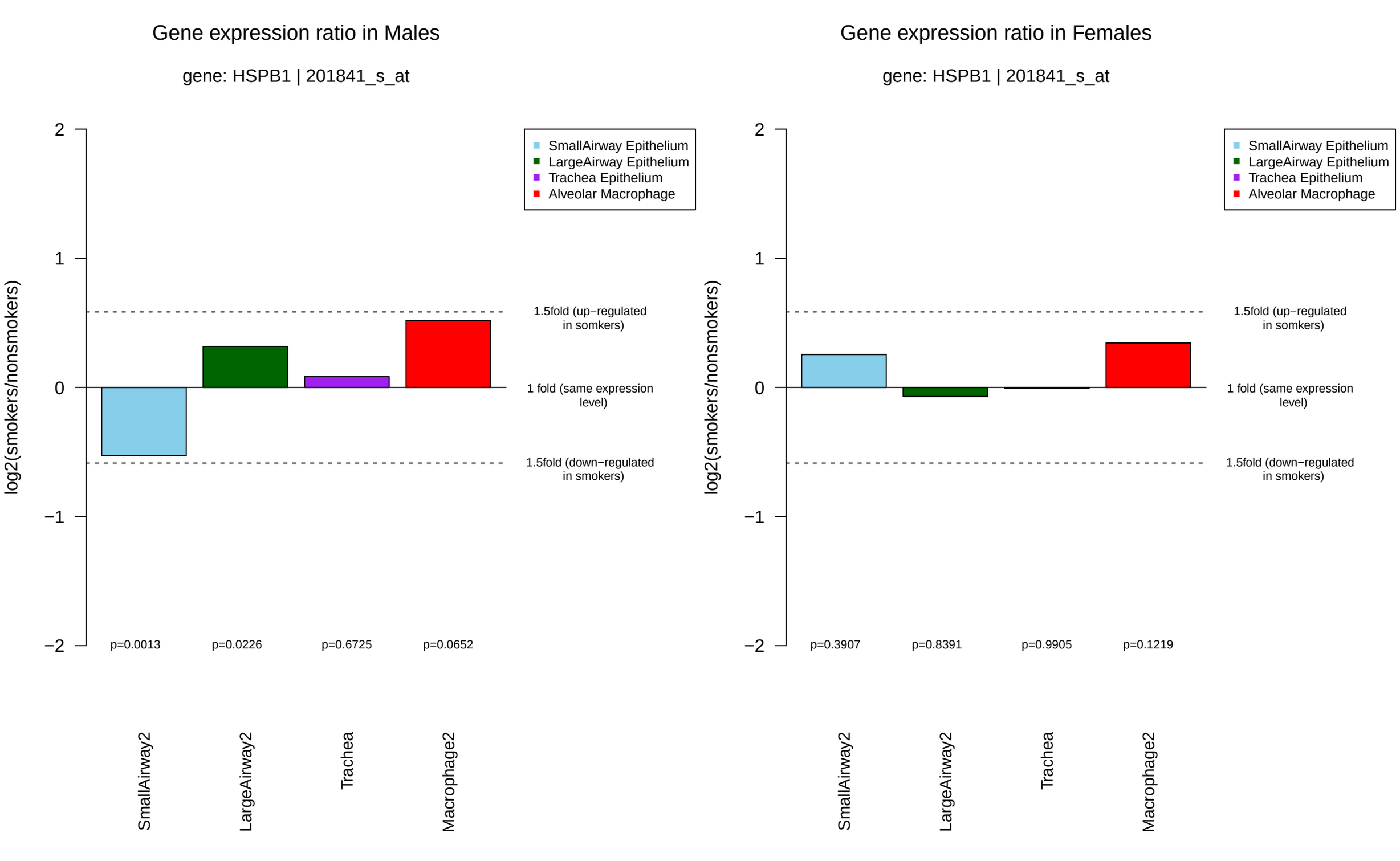
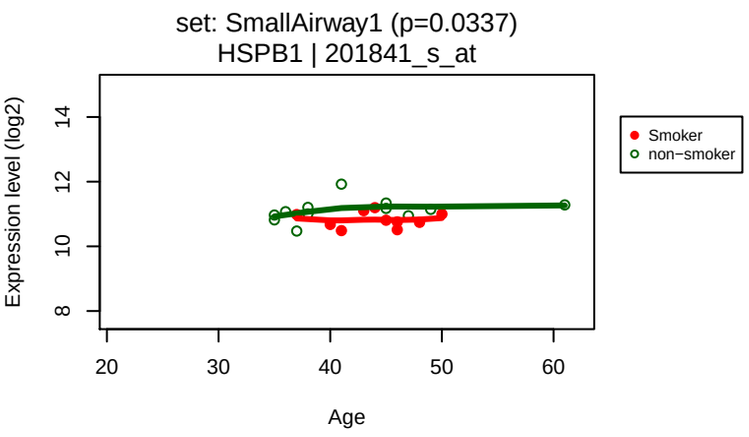
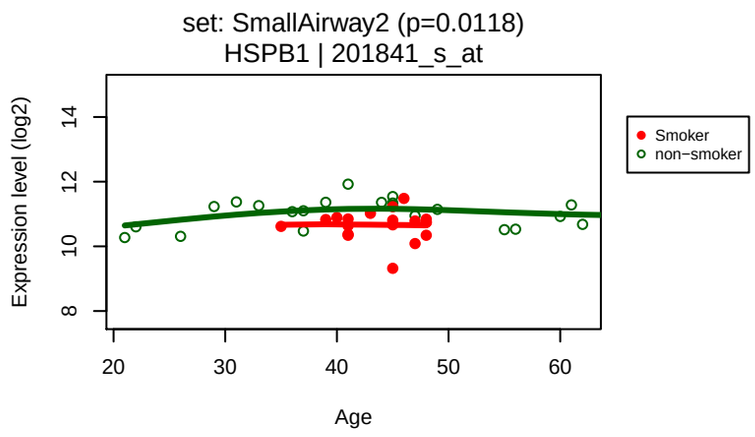
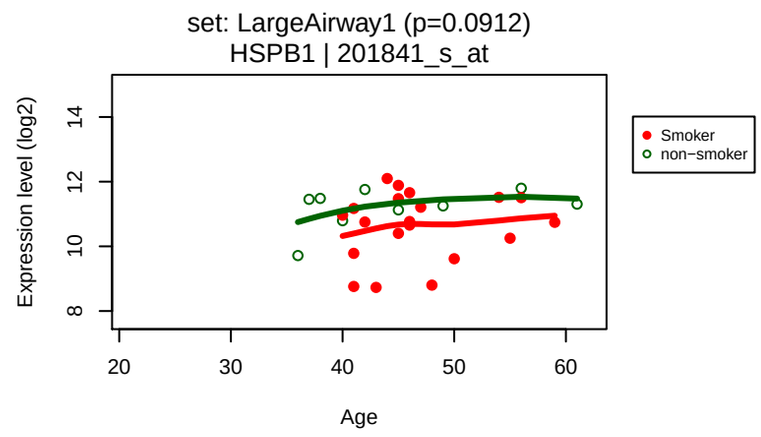

Gene expression of smokers and nonsmokers
gene: HSPB1 | 201841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0337
p=0.0118
p=0.0288
p=0.0912
p=0.187
p=0.8149
p=0.3776
p=0.0234
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

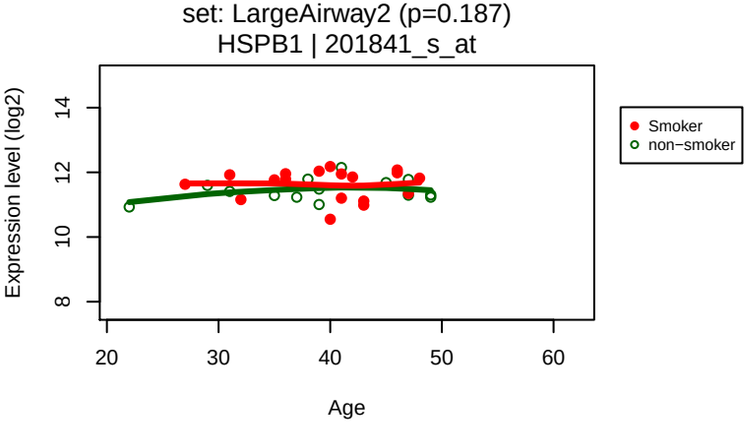

Gene expression ratio (smokers/nonsmokers)
gene: HSPB1 | 201841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0337
p=0.0118
p=0.0288
p=0.0912
p=0.187
p=0.8149
p=0.3776
p=0.0234
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

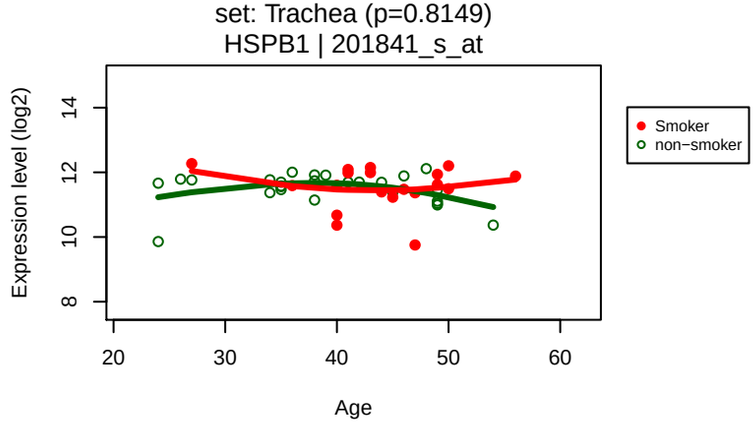

Gene expression ratio in Males
gene: HSPB1 | 201841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0013
p=0.0226
p=0.6725
p=0.0652
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB1 | 201841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3907
p=0.8391
p=0.9905
p=0.1219
SmallAirway2
LargeAirway2
Trachea
Macrophage2

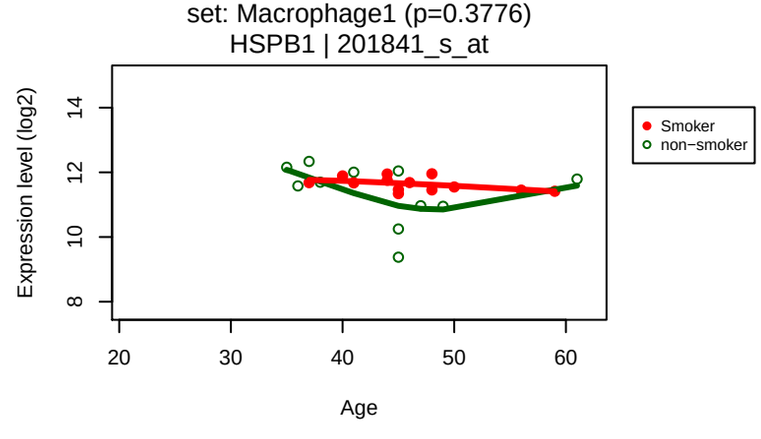

set: SmallAirway1 (p=0.0337)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker

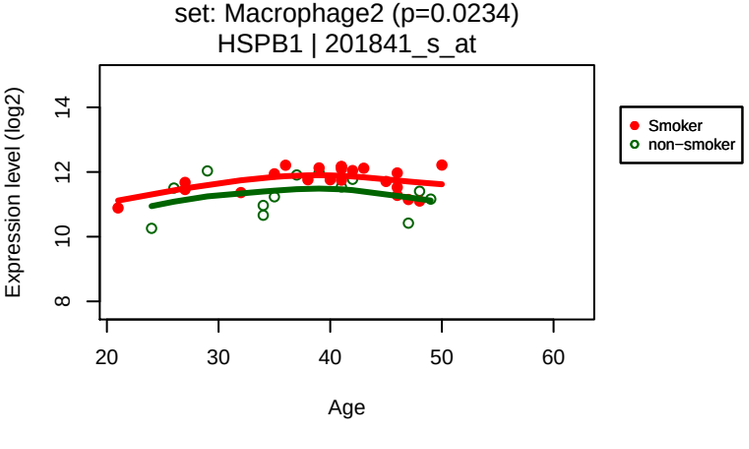

set: SmallAirway2 (p=0.0118)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker

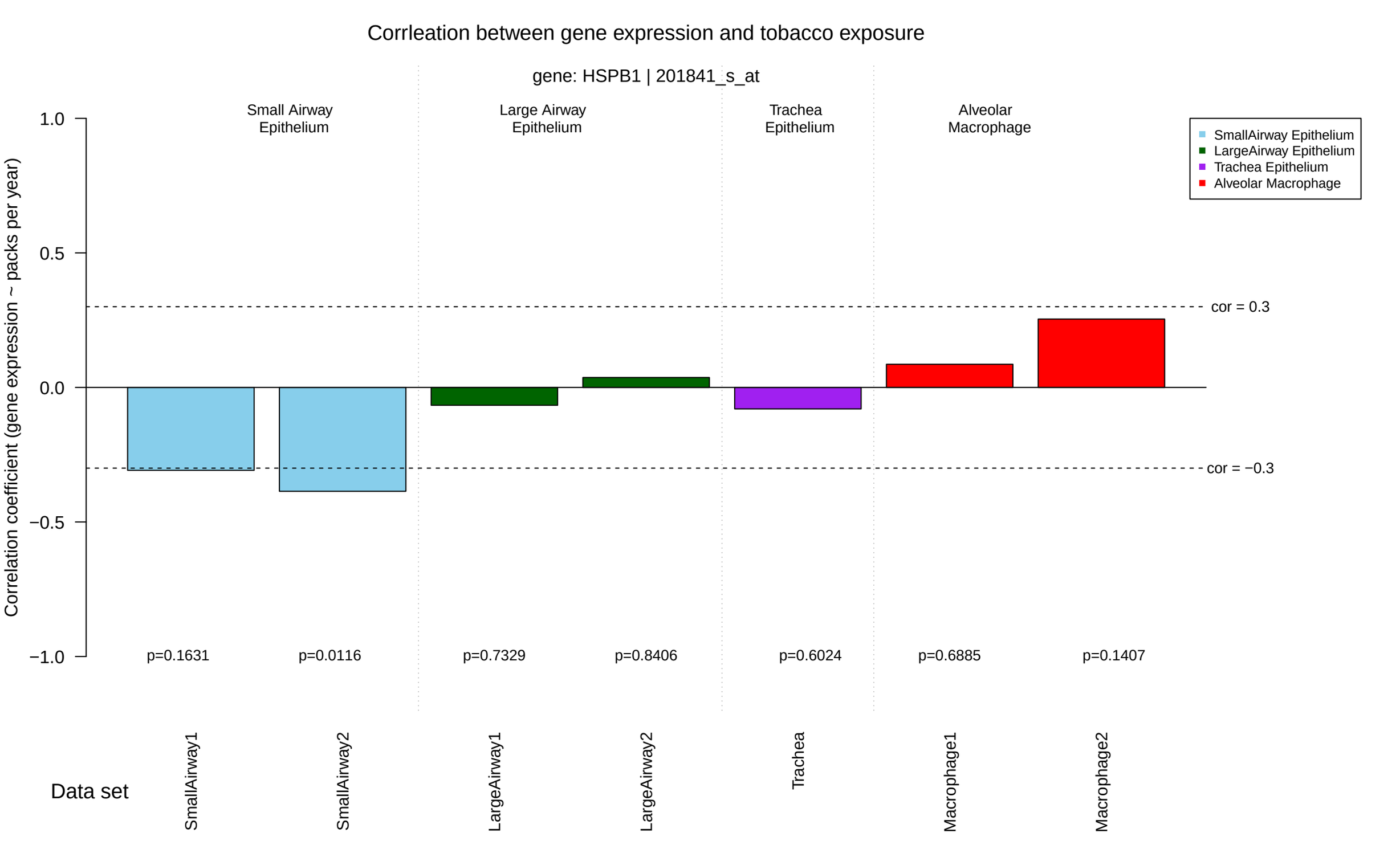

set: LargeAirway1 (p=0.0912)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.187)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8149)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3776)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0234)
HSPB1 | 201841_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB1 | 201841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1631
p=0.0116
p=0.7329
p=0.8406
p=0.6024
p=0.6885
p=0.1407
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2